Practical 3b: Multicast Delegate

AIM : To demonstrate a multicast delegate in C# that holds references to multiple methods and invokes them all in sequence.

PROGRAM CODE

```
using System;

namespace MulticastDelegate
{
   delegate void PrintDelegate(string message);

   class Printer
   {
      public static void PrintUpperCase(string msg)
      {
         Console.WriteLine($"UPPER: {msg.ToUpper()}");
      }

      public static void PrintLowerCase(string msg)
      {
         Console.WriteLine($"lower: {msg.ToLower()}");
      }

      public static void PrintReverse(string msg)
      {
         char[] chars = msg.ToCharArray();
         Array.Reverse(chars);
         Console.WriteLine($"Reversed: {new string(chars)}");
      }
   }

   class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("=== Multicast Delegate Demo ===");

         PrintDelegate del = Printer.PrintUpperCase;
```

```
        del += Printer.PrintLowerCase;
        del += Printer.PrintReverse;

        Console.WriteLine("Invoking multicast delegate:");
        del("Hello World");

        Console.WriteLine("\nRemoving PrintLowerCase:");
        del -= Printer.PrintLowerCase;
        del("Hello World");

        Console.ReadKey();
    }
  }
}
```

OUTPUT

```
=== Multicast Delegate Demo ===
Invoking multicast delegate:
UPPER: HELLO WORLD
lower: hello world
Reversed: dlroW olleH
```

```
Removing PrintLowerCase:
UPPER: HELLO WORLD
Reversed: dlroW olleH
```